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The Economics of Equity and Affordability in Residential Water Pricing

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In this review, I explore the economic dimensions of equity and affordability in urban residential water pricing. I summarize the growing literature on water affordability and discuss its relationship to more common studies of distributional issues in water demand. I also explore how utilities use various rate structures, such as increasing block rates, to address affordability concerns, emphasizing empirical evidence on the ineffectiveness of these rate structures in achieving equity goals. Moreover, I discuss the design of customer assistance programs in alleviating burdens on low-income households and summarize the scant evidence of the effectiveness of these programs. Ultimately, I highlight the need for more empirical evaluations of equity-focused programs and a greater focus on the distributional consequences of water management strategies to ensure equitable access to this essential resource.

Keywords: Water; distributional effects; equity; affordability.

JEL Classification: Q54, R23, R11

1. Introduction

Ensuring affordable access to clean and reliable water services is recognized as a fundamental human right (UN General Assembly 2010). However, the rising costs of maintaining and upgrading aging water infrastructure, coupled with the need for sustainable water management in the face of climate change, create significant challenges for water utilities and policymakers that need to balance service costs with equity goals (Allaire and Dinar 2022; Cardoso and Wichman 2022; Patterson *et al.* 2023; EPA 2024). In this review, I address a core tension between pricing water efficiently to reflect its value and encourage conservation, while simultaneously safeguarding access for vulnerable populations and mitigating disproportionate expenditure burdens.

Specifically, I focus on economic issues that characterize water affordability and the equity implications of residential water provision. The primary goal is to summarize key findings from the broad literature on water demand and the smaller, but growing, literature on water affordability, drawing insights from studies in both wealthy countries and low- and middle-income countries (LMICs). Because water affordability concerns are becoming increasingly salient in developed countries (EPA 2024), I focus the discussion on the United States, although many of the broader implications are applicable across different contexts.[1,2] By synthesizing existing research, this review aims to provide a clearer understanding of the economic principles at play and the practical challenges in achieving equity goals in water management.

This review comes at an important time in water policy. Many cities, states, and national governments are exploring options to implement water affordability programs. At the same, much of our water infrastructure (e.g., water treatment facilities, water mains, lead service lines) is in dire need of replacement (American Water Works Association 2012), which is pressurizing concerns about the affordability of water and wastewater service. Part of this need for new investment is driven by decades of underfunded water systems and underpriced water service. Historical disinvestment leads to a conundrum: water is often priced too cheaply to cover operating costs today without regard to future infrastructure needs, yet much of the world's population, even in wealthy countries, struggles to pay for basic water services and maintain reliable access. Rising infrastructure costs and climate-induced scarcity exacerbate these concerns, but economists' preferred market solution (i.e., full-cost pricing) amplifies equity concerns. What are the most effective ways to allow prices to reflect the value that water contributes to society, and encourage its efficient and sustainable management, without unduly burdening the most vulnerable members of society?

Within this review, I establish several key facts about water affordability and distributional issues, supported by both economic theory and empirical evidence.

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[1]This review complements several previous reviews that focus on water affordability, pricing, and access issues in LMICs. See, for example, Cook *et al.* (2020a, 2020b).

[2]The focus of this is on urban residential households with access to centralized piped water systems. I acknowledge that many rural, indigenous, and tribal communities, both in the US and globally, lack such access and often face significantly higher costs and reduced reliability when securing drinking water (Wheeler *et al.* 2025). These circumstances represent an important but distinct dimension of residential water access and equity that is outside the scope of this review.

These facts encompass our current understanding of water affordability, which, unlike standard regressivity analysis, is grounded in ethical considerations of the "ability to pay" for an essential resource (Herrington *et al.* 2003; EPA 2024) and necessitates normative judgments about acceptable expenditure burdens relative to income (National Academy of Public Administration 2017; Teodoro 2018; Cardoso and Wichman 2022). Recent evidence suggests that nearly 10% of US households face unaffordable water services, defined as spending more than 4.5% of their income on water and wastewater bills (EPA 2024). This finding highlights the growing concern over water affordability, particularly among low-income households, who often bear a disproportionate burden of water costs relative to their income (Cardoso and Wichman 2022; Patterson *et al.* 2023).

Further, I summarize the extensive literature on water demand as it relates to both income and price elasticities, with a focus on what types of households are most price-elastic, as well as income-specific responses to non-price conservation policies. It is well known that water demand is inelastic with respect to income, with meta-analyses of hundreds of empirical estimates indicating that income elasticities typically fall within the range of 0.1 to 0.5 (Arbuéz *et al.* 2003; Dalhuisen *et al.* 2003; Havranek *et al.* 2018). It is also well known that water demand is price-inelastic, with average estimates ranging from −0.3 to −0.5 (Sebri 2014; Dalhuisen *et al.* 2003; Espey *et al.* 1997). However, I highlight the important finding that lower-income households generally exhibit higher price elasticities compared to their higher-income counterparts (e.g., Agthe and Billings 1987; Renwick and Archibald 1998; Hajispyrou *et al.* 2002; Groom *et al.* 2008; Szabo 2015; Wichman *et al.* 2016; Wietelman *et al.* 2024). This finding suggests that lower-income households adjust their consumption more in response to price changes, which increases their exposure to distributional concerns from water pricing policies. Additionally, I examine the interaction of non-price conservation policies with equity, noting that while behavioral responses to measures like outdoor watering restrictions might be similar across income groups (Wichman *et al.* 2016), the ultimate distributional consequences can differ due to varying consumption patterns and capacities for conservation (Renwick and Green 2000; Rachunok and Fletcher 2023; Wichman 2023).

Another thrust of research on water equity focuses on the degree to which water rate structures can alleviate affordability burdens. I elucidate the rationale behind the adoption of various pricing mechanisms by water utilities, particularly those motivated by distributional concerns. This discussion focuses largely on increasing block rates (IBRs), which are frequently implemented with the intention of making water more affordable for low-income households by charging lower prices for initial consumption blocks designed to cover basic needs (Agthe and Billings

1987; Barberán and Arbuéz 2009; Whittington 1992). However, the empirical evidence of whether IBRs are effective at achieving equity goals is not promising (Barberán and Arbuéz 2009; Dahan and Nisan 2007; Ruijs *et al.* 2008; Whittington *et al.* 2015; Andres *et al.* 2019). These rate structures do not effectively target subsidies to poor households, primarily because consumption is not a strong proxy for income, with larger, potentially lower-income households sometimes facing higher per-person costs (Barberán and Arbuéz 2009; Smith 2022; Levinson and Silva 2022). Furthermore, I also discuss the economic implications of a more direct and less distortionary approach to addressing equity goals within rate structures, like lump-sum transfers or progressive fixed fees (Barde and Lehmann 2014; Reynaud 2016; Wichman 2024; Borenstein *et al.* 2021). Simpler rate structures, possibly combined with targeted assistance programs, might offer more desirable equity outcomes (Wietelman *et al.* 2024).

I also review the existing literature on customer assistance programs (CAPs), which aim to alleviate the burden of water costs on low-income households more directly. While these programs are designed to improve access to affordable water services, their effectiveness in practice is often uncertain and lacks rigorous empirical evaluation (Cook *et al.* 2020b). Despite significant efforts in designing and targeting these programs, there is a remarkable lack of empirical evidence regarding their overall success in alleviating affordability burdens and influencing water use in both low- and middle-income countries (LMICs) and wealthy nations. A better understanding of what does and does not work is critical for informing future policy decisions, especially as many cities, states, and national governments are exploring options to implement water affordability programs (EPA 2024).

Finally, I conclude with a discussion of the future research directions that can help address the gaps in our understanding of water equity and affordability. What becomes clear after evaluating the existing literature is that we know a great deal about how prices, income, and conservation programs affect water use. However, we know considerably less about the distributional consequences of water management policies and the effectiveness of programs designed to alleviate concerns about water affordability. This review serves as a call-to-action for more empirical evaluations of equity-focused water policies and programs, to ensure equitable access to this essential resource.

2. Water Affordability

The concept of water affordability is distinct from standard economic analysis of regressivity by its grounding in ethical considerations of the “ability to pay” for an essential resource (Herrington *et al.* 2003; EPA 2024). Water affordability is predicated on the notion of water as a social good (i.e., a good the confers public benefits) and, in many perspectives, a human right (UN General Assembly 2010; Beecher 2020). This ethical underpinning suggests that all individuals, regardless of income, should be able to access basic water services for drinking, sanitation and hygiene without facing undue hardship (Raucher *et al.* 2019; Patterson *et al.* 2023; EPA 2024).

Unlike more standard analyses of economic incidence that examine distributional impacts of water expenditures across the income distribution, water affordability specifically addresses whether the poorest households can meet their essential needs without sacrificing other necessities (EPA 2024; Teodoro 2018). This distinction leads to discussions and implementation of policies such as increasing block tariffs (IBTs) with lower rates for initial consumption blocks, "lifeline" rates, and customer assistance programs (CAPs), all designed with the intention of safeguarding access to a minimum quantity of water at an affordable cost (Groom *et al.* 2008; Herrington *et al.* 2003).

Furthermore, affordability metrics used to evaluate water affordability must incorporate normative judgments about what constitutes an acceptable burden relative to income and essential consumption (National Academy of Public Administration 2017; Teodoro 2018; Cardoso and Wichman 2022; EPA 2024). Expenditure ratios that define a percentage of household income spent on water (e.g., 4.5% of household income) reflect societal values about the affordability of water service (Nemati and Schwabe 2023). Although economists more commonly undertake positive analyses to evaluate the incidence of water service costs across the income distribution, the specific focus on defining and measuring affordability against a basic needs standard and income levels distinguishes it from typical distributional analyses.[3]

2.1. *A brief history of water affordability in the US*

The history of water affordability in the United States is linked to the Environmental Protection Agency's (EPA) efforts to assess the financial capability of communities, particularly small drinking water systems, when faced with new regulatory requirements under the Clean Water Act (CWA) (EPA 1997 2024; National Academy of Public Administration 2017). Initially, the EPA primarily used Median Household Income (MHI) as a key indicator, with a 4.5% of MHI threshold for combined water and sewer bills being considered a reasonable upper limit (with a 2% threshold for water service only). This metric helped the EPA

[3]This approach to measuring affordability, while potentially flawed, is common in other domains for essential needs (e.g., housing, energy, food assistance).

conduct technology affordability assessments to determine how small systems could comply with water quality standards without imposing undue financial burden on their communities. The underlying goal was to ensure that environmental regulations were achievable without generating widespread affordability challenges.

Over time, the 4.5% of MHI threshold for combined water and sewer bills (CWSBs) gained prominence as a benchmark for evaluating water affordability more broadly. This threshold, although widely used in practice and literature to compare expenditure burdens, lacked a strong theoretical or empirical basis. Many researchers and practitioners have argued that MHI is an inadequate metric for truly understanding household-level affordability, as it fails to capture the financial hardships of the lowest-income ratepayers within a community (National Academy of Public Administration 2017; Teodoro 2018; Cardoso and Wichman 2022; Nemati and Schwabe 2023; Patterson *et al.* 2023). Consequently, there has been a growing call for more nuanced and comprehensive approaches to measuring water affordability that move beyond simplistic MHI-based thresholds and better reflect the diverse financial realities of households. A 2017 study recommended that the EPA improve its Financial Capability Assessment (FCA) framework to address these limitations (National Academy of Public Administration 2017), and EPA has moved towards applying the 4.5% water-expenditure-to-income threshold at the *household* level (EPA 2024).

2.2. *Current trends in water affordability*

Even in wealthy countries such as the United States, water affordability is an increasingly pressing concern. This fact is underscored by a recent EPA report to Congress that estimates that 9.2% of US households ($\sim$12 million households) lack affordable access to water services, where "lack of affordable water" is defined as expenditures on an essential level of water and sewer service exceeding 4.5% of household income (EPA 2024).[4] EPA's estimates follow methodologies from Cardoso and Wichman (2022) and Patterson *et al.* (2023), which focus on smaller geographies. In those primary studies, Cardoso and Wichman (2022) find that around 10% of US households face affordability concerns when defined as water expenditures on essential water services (defined as 50 gallons per person per day) exceeding 4.5% of annual income. Patterson *et al.* (2023) find that 17.1% of households experience unaffordable water services based on a threshold of 4.6% of household income. Several studies (e.g., Teodoro 2018; Teodoro and Saywitz

[4] Using a lower affordability threshold of 3%, EPA finds 14.6% of US households (or 19.2 million households) face unaffordable water and sewer service.

2020) also indicate increasing water affordability challenges when using alternative benchmarks like comparing bills to hours worked at the minimum wage.

These studies consistently demonstrate that low-income households bear a disproportionately high burden of water costs relative to their income. EPA (2024) reports that over 75% of households in the lowest income quintile face unaffordable bills at a 3% threshold. Cardoso and Wichman (2022) estimate that the average household in the lowest income decile spends an average of 6.8% of their annual income on essential water and sewer services. A key takeaway across these analyses is the importance of using income data on a fine geographical scale (e.g., household) to accurately assess affordability burdens, as community-level median income can mask the impacts on the poorest households. Finally, regional focus and other socioeconomic factors influence affordability burdens. Patterson *et al.* (2023) found higher levels of unaffordable water services in the Southeast and Midwest compared to the Southwest, even with comparable bill amounts, due to a larger share of households with low incomes.[5] Further, Teodoro and Saywitz (2020) also found that water and sewer services tend to be more affordable in the Western US and that greater income inequality within a community is associated with increased affordability challenges. Cardoso and Wichman (2022) found that US geographies with more Black households faced a larger affordability burden, but this result did not exist for geographies with larger concentrations of Hispanic residents.

Water affordability trends are driven predominantly by the increasing costs associated with providing clean and reliable water services (Herrington *et al.* 2003; EPA 2024; Patterson *et al.* 2023). This rise in costs stems from several factors, including the critical need for maintaining and upgrading aging water infrastructure, which requires substantial capital investment (Allaire and Dinar 2022; Patterson *et al.* 2023). More stringent environmental and health regulations often necessitate costly upgrades to treatment processes and infrastructure (EPA 2024). Rising operational costs, such as those related to energy for pumping and the purchase of treatment chemicals, also contribute to the upward pressure on water rates (EPA 2024). Further, El-Khattabi *et al.* (2023) found that municipally owned water utilities tend to have lower bills, while those that primarily purchase their water typically charge higher rates. These increasing costs, when passed on to consumers through water bills, can disproportionately affect households with limited incomes, making water services less affordable.

[5]This finding is consistent with Luby *et al.* (2018) who found that water prices are lower in drier regions, largely reflecting the fact that costs are driven by infrastructure needs rather than scarcity, and water infrastructure is generally newer in Western cities.

In addition to the rising cost of service, household income levels and the structure of water rates are key determinants of affordability. Stagnant or low incomes, particularly among the poorest households, mean that water expenditures constitute a larger percentage of their already strained budgets (EPA 2024; Cardoso and Wichman 2022). How water utilities design their rates, specifically the balance between fixed service charges and volumetric consumption charges, is also a significant contributing factor (Allaire and Dinar 2022; Schmidt and Lewis 2017). Higher fixed charges can place a greater burden on low-income households regardless of water consumption levels (Schmidt and Lewis 2017; Wichman 2024).[6]

Water affordability is a salient issue in countries beyond the United States, as well. In Southern Spain, research indicates that the cost of basic water needs is relatively higher for lower-income households, driven by potential inequities in tariff structures (García-Valiñas *et al.* 2010). An earlier study focusing on OECD countries suggested that water affordability for low-income populations is either a current concern or could become one without appropriate policy interventions (Herrington *et al.* 2003). In many developing countries, despite the use of non-linear tariffs intended to aid the poor which are often poorly targeted, affordability and access remains a significant challenge. This is true especially as utilities move towards pricing that reflects the full cost of service provision (Whittington *et al.* 2015; Komives *et al.* 2005; García-Valiñas *et al.* 2010). In Europe, however, there is some evidence that a move towards full cost-recovery would not dramatically exacerbate affordability issues (Reynaud 2016).

2.3. *Relationship between affordability and regressivity*

Water affordability and regressivity of water expenditures are distinct concepts. However, unaffordable water often implies regressivity. Regressive water bills refer to the case where the proportion of a household's income devoted to water expenditures decreases as household income increases (Cardoso and Wichman 2022; Timmins 2002a). This is distinct from affordability, which is often framed as whether the total cost of water services places an undue burden on a household's budget, potentially impacting their ability to pay for other necessities (Herrington *et al.* 2003; Teodoro 2018). Generally speaking, water bills are regressive. For instance, Cardoso and Wichman (2022) demonstrate regressivity by showing that the lowest income decile in the US spends a substantially higher percentage of their income (6.8%) on essential water and sewer services compared to median-income households (around 1%).

[6]The interaction of water rate structures and affordability is discussed in more detail in Sec. 4.

Water bills often act as a regressive levy because the demand for essential water is relatively inelastic, particularly at low consumption levels (Espey *et al.* 1997; Dalhuisen *et al.* 2003). Lower-income households spend a larger fraction of their income on necessities, including a basic amount of water, and thus are more significantly impacted by water charges as a proportion of their budget (Beecher 2020; Timmins 2002a). Timmins (2002a) found that water charges are a more regressive revenue source for municipalities compared to property or sales taxes. Furthermore, fixed charges associated with water service exacerbate this regressivity, as these costs are common to all customers and, by definition, represent a larger relative burden for households with lower incomes (Beecher 2020; Wichman 2024).[7] Although affordability policies might aim to ensure water affordability for all income levels through measures like lifeline rates or rebates, the underlying structure of water tariffs often results in a regressive distribution of costs (Wietelman *et al.* 2024). To aid in understanding the role of regressivity, economists often analyze income elasticities of water demand to understand how consumption changes with income and how pricing structures contribute to or mitigate the regressivity of water bills, which is the focus of the subsequent section.

3. Income and Water Demand

Historically, the focus on the relationship between water use and income has been concentrated in studies of water demand. These studies have estimated income elasticities of demand, to measure how water use changes with changes in income, as well as price elasticities of demand. Many of these studies also estimate price responsiveness across different income groups, as well as other demographic characteristics, to understand how changes in water prices may differentially impact high- vs. low-income households. A complementary literature has also focused on who bears the burden of non-price conservation policies, such as water restrictions during drought, which additionally capture unequal burdens in demand-side management. In this section, I summarize the primary takeaways from this literature as it relates to equity and affordability concerns.

3.1. *Does water use vary with income*?

Research on water demand overwhelmingly indicates that water demand is income inelastic. This means that when household income changes, the demand for water changes by a proportionally smaller amount. Meta-analyses of hundreds of

[7] Wichman (2024) makes the point that the fixed charges on water bills operate similar to a lump-sum tax and are likely the most regressive form of revenue raised by municipalities.

empirical estimates have found income elasticities to be positive but substantially less than one, typically falling within the range of 0.1 to 0.5 (Arbuéz *et al.* 2003; Dalhuisen *et al.* 2003; Havranek *et al.* 2018). The primary finding from Havranek *et al.* (2018) is that the income elasticity of water demand is around 0.26, but after correcting for publication and endogeneity biases, the mean underlying elasticity is estimated to be approximately 0.15 or smaller, which suggests that income is only weakly associated with water demand. If income were to double, water consumption would be expected to rise by only 15%.[8]

Havranek *et al.* (2018) find that several factors drive differences in income elasticity estimates. For example, the presence of increasing block rates are associated with relatively small income elasticities. The regional focus of the study, the granularity of the data used (e.g., annual, monthly) in the underlying study, and how the study accounts for weather can systematically affect the estimated income elasticities. Methodological choices in econometric modeling and the type of data employed (e.g., panel, cross-sectional, aggregated or household-level) also contribute to the variation in estimates.

The relationship between household size and income is an important consideration when analyzing water demand because the correlation between these two factors can influence consumption patterns (Cardoso and Wichman 2022; Whittington *et al.* 2015). Although it is often assumed that higher incomes correlate with higher water use, this relationship can be complicated by household size. For instance, low-income households may have larger families (e.g., more likely to have multi-generational households), potentially leading to higher overall water consumption despite lower per capita use (Whittington *et al.* 2015). Conversely, in other contexts, average household size might increase with income (e.g., two working spouses are more likely to have larger household sizes and larger incomes than a single-person household). Additionally, within-household economies of scale in household water use suggests that as household size increases, water consumption does not rise proportionally (Arbuéz *et al.* 2003; Binet *et al.* 2014; Arbuéz *et al.* 2010; Schleich and Hillenbrand 2009). These efficiencies occur because certain uses like domestic cleaning, cooking or lawn-irrigation have public good characteristics and do not scale directly with the number of residents (Arbuéz *et al.* 2010). Therefore, understanding the interaction between household size, income and economies of scale is critical for accurately modeling and predicting residential water demand.

[8]However, income elasticities are defined as a percentage change in water use in response to a *marginal* change in income, so this illustrative example is not well defined.

Although many studies have estimated income elasticities, rarely are these estimates the primary focus of the analysis. Further, there are not good natural experiments to causally identify the effect of changes in income within households or even within utility service areas. Instead, most studies rely on cross-sectional data to estimate income elasticities, which can be biased by omitted variables. Despite this, there are important questions that deserve further investigation. For example, are higher income households more likely to adopt water-saving technologies, such as high-efficiency appliances (e.g., Bennear *et al.* 2013) or water-efficient landscapes (e.g., Brelsford and Abbott 2021; Burkhardt *et al.* 2022)? On the other hand, high-income households may also be more likely to have larger and greener lawns, landscaped gardens, swimming pools, or greater demands for other water-intensive household goods.

3.2. *Does price responsiveness vary with income*?

Estimating consumer responsiveness to changes in price has been the predominant focus of water demand studies dating back nearly 100 years (Metcalf 1926). There is very little disagreement that water demand is price-inelastic, with several meta-analyses suggesting average values ranging from -0.3 in Sebri (2014), -0.4 in Dalhuisen *et al.* (2003) and -0.5 in Espey *et al.* (1997).[9] In this extensive literature, numerous studies have explored whether price sensitivity to vary across different income groups. A prevalent finding across these studies is that lower-income households generally exhibit higher price elasticities compared to their higher-income counterparts (Agthe and Billings 1987; Renwick and Archibald 1998; Hajispyrou *et al.* 2002; Groom *et al.* 2008; Szabo 2015; Wichman *et al.* 2016; Wietelman *et al.* 2024). This finding implies that when the price of water changes, households with lower incomes tend to adjust their consumption by a greater percentage than those with higher incomes. However, because higher-income households tend to consume more water overall, the absolute change in water use in response to price changes can be larger for higher-income households, even if their relative change is smaller.

The primary reason for this higher price sensitivity among lower-income households is often attributed to the fact that water expenditures constitute a larger fraction of their overall household budget (Howe and Linaweaver Jr. 1967; Arbuéz *et al.* 2010). Consequently, any increase in water prices can have a more substantial

[9]Interestingly, early studies of water demand with much cruder data and methods than what is typically used today found similar estimates: Metcalf (1926) estimated elasticities of around -0.65, Gottlieb (1963) estimated an average elasticity of -0.4, and Howe and Linaweaver, Jr. (1967) also found an average elasticity of -0.4, which is a weighted average of responsiveness of indoor and outdoor demands.

impact on their financial well-being, necessitating a more pronounced behavioral response in terms of reduced water usage. For example, Zapata (2015) found that low-income households in Ecuador were more responsive to changes in water prices compared to middle and high-income households. Wichman *et al.* (2016) found that lower-income households in North Carolina were more responsive to water price changes than high-income households. Similarly, Wietelman *et al.* (2024) document that the lowest-income households had the most elastic demand response to drought surcharges implemented by two Southern California water providers.

However, the relationship between income and price sensitivity is not strictly inverse, and substantial price responsiveness has also be estimated for higher-income households, particularly for certain types of water use. Mansur and Olmstead (2012) found that their "poor, small-lot" group had the most elastic outdoor demand, but also noted relatively larger indoor elasticities for "rich, big-lot" households, suggesting that income interacts with other factors like lot size and the nature of water use (indoor versus outdoor) to influence price responsiveness. Similarly, Baerenklau *et al.* (2014) found highest price-responsiveness among high-income households.[10] More recently, El-Khattabi *et al.* (2021) found that heavy water users, even among high-wealth households, exhibited a larger degree of responsiveness to price changes than light or moderate users in the same wealth category. These findings indicate that while the magnitude of response might differ, higher-income households are not necessarily immune to price signals and may adjust discretionary water consumption in response to price changes.

In general, there is a pattern that suggests that lower-income households are typically more sensitive to price, though it is important to recognize heterogeneity in price elasticities across all income levels. Factors such as the necessity of water for basic uses (Arbuéz *et al.* 2003), the presence of discretionary uses, household size (Arbuéz *et al.* 2010), and even consumers' perception of prices (Gaudin 2005; Binet *et al.* 2014) can contribute to variations in price responsiveness within and across income groups.

3.3. *Does the effectiveness of non-price conservation policies vary with income*?

During times of acute scarcity, water managers are often reluctant to let prices rise to reflect scarcity and to induce conservation, largely driven by distributional concerns since low-income households tend to be more sensitive to price increases (Renwick and Archibald 1998; Wichman *et al.* 2016). Instead, utilities often adopt

[10] Baerenklau *et al.* (2014) also found low-usage households to be relatively more price-responsive, which contrasts with Mansur and Olmstead (2012), Wichman *et al.* (2016) and El-Khattabi *et al.* (2021).

non-price conservation measures (such as outdoor watering restrictions, public education campaigns, and rationing). Although these conservation initiatives can also have distributional consequences, evidence of responsiveness to these measures appears to be shared more equitably across the income distribution. For instance, research in North Carolina found little observable heterogeneity in how various income groups reduced their water use in response to both voluntary and mandatory outdoor watering restrictions during drought (Wichman *et al.* 2016). This finding is in contrast to the general finding that lower-income households typically show a greater sensitivity to changes in water prices (Renwick and Archibald 1998; Mansur and Olmstead 2012; Wichman *et al.* 2016).

Although the behavioral response to non-price conservation measures exhibits limited variation across income groups, the ultimate distributional consequences may differ since different income levels correlate with different water consumption patterns and capacities for conservation (Renwick and Green 2000; Rachunok and Fletcher 2023; Wichman 2023). For example, when landscape watering restrictions are implemented, higher-income households, who tend to have larger lawns and more extensive landscaping, will likely experience more substantial absolute reductions in their water use compared to lower-income households with smaller or no yards (Renwick and Green 2000).

Additionally, the impact of water conservation measures on water expenditures can also vary by income. Rachunok and Fletcher (2023) point out that during droughts, non-price measures like curtailment can lead to greater reductions in water bills for high-income households who typically have more discretionary outdoor water consumption that can be curtailed without affecting essential needs. In contrast, lower-income households, who often have less discretionary use, might not achieve the same level of bill savings, and the required revenue adjustments by utilities could disproportionately affect their more constrained budgets (Rachunok and Fletcher 2023).

The discussion so far focuses primarily on mandatory command-and-control measures, but there is some evidence of heterogeneous effects of voluntary conservation measures, such as rebates, information campaigns, and behavioral nudges, across different income levels or user types often correlated with income (Maas *et al.* 2024). Behavioral nudges, including social comparisons, have emerged as a promising tool for encouraging water conservation (Ferraro and Price 2013; Ferraro and Miranda 2013; Brent *et al.* 2017; 2020). Some studies indicate that social comparison messages are most effective among households identified as high users (Ferraro and Price 2013) as well as high-income households as proxied by home value (Ferraro and Miranda 2013). Since higher-income households often tend to be less price sensitive and are frequently high users, this suggests that such norm-based strategies can effectively engage this demographic and serve as a useful complement to price measures for groups less responsive to price changes. Furthermore, Brent *et al.* (2017) show that social comparison messages may increase participation in existing conservation programs, potentially attracting less price-sensitive, high-use customers to initiatives like rebates.

Rebate and subsidy programs designed to incentivize the adoption of water-efficient technologies or turf-replacement can also be effective demand-side management policies (e.g., Brelsford and Abbott 2021; Sovocool *et al.* 2006; Burkhardt *et al.* 2022; Bennear *et al.* 2013; Price *et al.* 2014). However, these programs may be regressive in nature, as households with sufficient disposable income are typically better positioned to afford the upfront purchase of efficient products or to undertake the necessary investments (Maas *et al.* 2024). Bennear *et al.* (2013) focus on rebates for high-efficiency toilets and show that high-income and highly educated customers are more likely to receive rebates, although they would have likely purchased a high-efficiency toilet without the rebate, leading to low additionality of the program. Although some studies have analyzed the effectiveness of rebates for turf-grass replacement in the arid West (Brelsford and Abbott 2021; Burkhardt *et al.* 2022; Sovocool *et al.* 2006), very few have examined the distributional consequences of these programs across income groups. However, the people most likely to benefit from these programs are homeowners with green lawns.

Overall, the evidence suggests that income differences do not drive differences in responsiveness to non-price water conservation measures. Households across income levels generally tend to alter their behavior similarly when faced with such restrictions, although there is some evidence that high-income households are more responsive to behavioral nudges and more likely to benefit from rebates for water-efficient technologies. However, the distributional consequences of these policies can still be significant due to the differences in water use patterns and the ability to reduce consumption without impacting essential needs, which can correlate with income (Renwick and Green 2000; Rachunok and Fletcher 2023). Therefore, while non-price measures might be seen as more equitable in terms of initial response compared to price hikes, policymakers must still consider the aggregate implications across income groups.

4. Do Utilities set Rates to Improve Equity, and are they Effective?

4.1. *Drivers of utility rate setting*

Water utilities set prices by balancing several key objectives, including recovering the cost of service, promoting efficiency and conservation in water use, maintaining a stable revenue stream, and addressing equity concerns (American Water Works Association 2021; Hanemann 1997; Barberán and Arbuéz 2009;

Diakité *et al.* 2009; Reynaud *et al.* 2005; Wichman 2024).[11] To achieve these multifaceted and sometimes conflicting goals, utilities use a range of common pricing structures that are broadly categorized as two-part tariffs.[12] These rate structures can include uniform rates, where all water consumption is charged at the same volumetric price; IBRs, which impose progressively higher prices as consumption increases; and sometimes seasonal or time-of-use rates that reflect variations in supply costs or demand patterns (Allaire and Dinar 2022; Monteiro and Roseta-Palma 2011; Olmstead *et al.* 2007; Reynaud *et al.* 2005). More recently, utilities have begun adopting "budget-based rates" (BBRs), which are effectively individualized IBRs in which a household's "budget" is assigned based on household and property characteristics and weather (Baerenklau *et al.* 2014; Smith 2022; Wietelman *et al.* 2024).

The selection of a particular rate structure and the determination of specific price levels are influenced by a variety of factors. These include the utility's operational expenses, infrastructure maintenance and investment needs, characteristics of the customer base such as income distribution and consumption habits, prevailing environmental conditions, and relevant regulatory requirements and local ordinances (see, e.g., Allaire and Dinar 2022; Gaudin 2005; Olmstead *et al.* 2007; Maas *et al.* 2024; Hanemann 1997). Utilities often face trade-offs when trying to reconcile these different objectives, such as the balance between ensuring revenue stability and incentivizing conservation through pricing signals.

Equity considerations are a prominent driver of how utilities design and implement water pricing policies (Barberán and Arbuéz 2009; Herrington *et al.* 2003; Reynaud *et al.* 2005). Recognizing that access to clean and safe water is often considered a fundamental human right and a social good that confers both private and public benefits (Beecher 2020; Hajispyrou *et al.* 2002; Herrington *et al.* 2003), utilities and policymakers are particularly concerned with ensuring that water services remain affordable for all customers. This fact is especially true for low-income households, who may spend a larger proportion of their income on water and thus face a disproportionate burden from higher water bills (Whittington 1992; García-Valiñas *et al.* 2010; Cardoso and Wichman 2022). To address these equity concerns,

[11]Economists often favor marginal cost pricing for water because it is seen as a way to achieve economic efficiency by setting the price of water equal to the cost of supplying an additional unit (Coase 1946; Howe and Linaweaver, Jr. 1967; Hanemann 1997; Olmstead 2010). Many economists argue that water rates should reflect the long-run marginal cost (LRMC) of provision to account for both current and future costs, as cheaper water sources are typically developed first, leading LRMCs to be larger than short-run marginal costs (Renzetti 1992; Hanemann 1997; Olmstead 2010).

[12]A two-part tariff typically consists of a fixed charge that a customer pays regardless of water consumption and a volumetric charge that is applied per unit of water consumed. These rate structures can be traced back, at least, to Coase (1946).

utilities often adopt rate structures that attempt to mitigate the financial impact on vulnerable populations.

One of the most common approaches to incorporating equity into water pricing is the use of IBRs (Agthe and Billings 1987; Barberán and Arbuéz 2009; Whittington 1992; Olmstead *et al.* 2007; Whittington *et al.* 2015; Wichman 2024).[13] These tariff structures are designed with the intention of making water more affordable for low-income households by charging smaller marginal prices for initial blocks of water consumption — sometimes referred to as "lifeline" blocks — which are intended to cover basic needs (Boland and Whittington 2000; Groom *et al.* 2008; Whittington 1992).[14] The expectation is that households with lower incomes typically consume less water and will thus primarily fall into these lower-priced tiers. Conversely, higher levels of water consumption, often associated with wealthier households and discretionary uses like lawn watering or swimming pools, are priced at progressively higher rates. This tiered structure is intended to subsidize essential water use for the poor through revenues generated from higher consumption by wealthier users (Diakité *et al.* 2009; Groom *et al.* 2008; Nauges and Whittington 2017). In a recent paper, Randriamaro and Cook (2024) find that among the largest 225 US cities, water rates are more redistributive in cities with higher income inequality, suggesting that utilities are more likely to use rates to achieve equity goals in cities with greater income disparities.

4.2. *Are equity-focused rate structures misguided*?

A general principle in economics is that equity and efficiency goals can (and should) be separated.[15] Equity goals should be achieved through lump-sum transfers as to not introduce allocative inefficiencies by allowing prices to deviate from marginal costs of provision. Another general principle is that to achieve any number of policy goals, a policymaker needs as many instruments as there are goals (Tinbergen 1956; Wheeler *et al.* 2025). It may strike economists as odd, then, that water utilities often use a single marginal instrument (the volumetric price of water) in an attempt to achieve equity goals.

Utilities often discount water prices below marginal costs to shield customers from the full cost of service provision (Timmins 2002a,b; Olmstead *et al.* 2007; Allaire and Dinar 2022; Wichman 2024). These discounts occur because of political

[13]There is a parallel literature that explores the rationale and effectiveness of increasing-block rates in promoting equity in electricity pricing as well (see, e.g., Borenstein 2012 2016; Borenstein *et al.* 2021; Levinson and Silva 2022).

[14]Some rate structures might actually provide an allocation of water at zero marginal price notionally intended to cover basic needs (Szabo 2015; Beecher 2020; Wichman 2024).

[15]Indeed, economics students typically learn the first and second fundamental theorems of welfare economics within the first few weeks of graduate school.

pressure to keep rates low and to provide affordable access to a basic necessity. From a theoretical perspective, pricing water below its marginal cost of provision provides an implicit subsidy for each unit of water consumed (Timmins 2002a; Szabo 2015), which occurs because the revenue generated by these lower prices does not fully cover the expense of providing that additional water. This subsidy often requires the utility to cover those costs through general tax revenues or by foregoing necessary infrastructure maintenance and upgrades. This underpricing also sends an incorrect price signal to consumers, potentially leading to overconsumption compared to the economically efficient level that would result from pricing at marginal cost.

Additionally, the use of IBRs (or similar rate nonlinear rate structures) to achieve equity goals is often seen as a "second-best" solution compared to more direct and less distortionary methods of income redistribution (Barde and Lehmann 2014; Reynaud 2016; Wichman 2024). Although the intuitive appeal of IBRs lies in the idea of subsidizing essential water use for low-income households through lower prices in initial blocks and charging higher prices to higher-income households, this approach is a crude way to target subsidies to low-income customers because income and water-use are only weakly correlated (Whittington *et al.* 2015; Levinson and Silva 2022).[16] Furthermore, by allowing marginal prices to vary across uses and households, IBRs introduce price distortions that can generate allocative efficiency costs, as consumers face different prices for the same good, potentially leading to suboptimal consumption decisions (Borenstein 2012 2016; Nauges and Whittington 2017; Schoengold and Zilberman 2014; Wichman 2024). In theory, direct transfers or progressive fixed fees are often considered more efficient instruments for addressing equity concerns without distorting the price signal for water consumption, allowing prices to be set closer to marginal cost to achieve efficiency goals (Beecher 2020; Borenstein *et al.* 2021; Wichman 2024; Wietelman *et al.* 2024). The justification for using IBRs for equity often arises in situations where such direct redistribution mechanisms are impractical or politically infeasible (Barde and Lehmann 2014; Groom *et al.* 2008; Reynaud 2016).

4.3. *Are equity-focused rate structures effective*?

Empirical evidence regarding the effectiveness of different water rate structures in achieving equity goals is not particularly favorable. While IBRs are frequently implemented with the intention of enhancing equity by providing lower prices for essential consumption levels, studies have yielded unencouraging results regarding

[16] However, Szabo (2015) does find that the implicit subsidies within nonlinear rate structures in South Africa act as lump-sum transfers to households, suggesting that the theoretical distinction between fixed and volumetric subsidies may not be empirically relevant.

their actual impact on income redistribution and affordability. For instance, Barberán and Arbuéz (2009) analyzed the IBR structure in Zaragoza, Spain, and identified an equity concern where larger households, which may include lower-income families, faced a higher per-person cost for meeting basic water needs compared to smaller households due to the tariff. Based on this analysis, they proposed a new rate design aimed at ensuring that all individuals, regardless of household size, could meet their basic water needs at the same cost. Dahan and Nisan (2007) also found that larger and often poorer households in Jerusalem pay higher average prices when block sizes are not adjusted for household size. Similarly, Ruijs *et al.* (2008) examined the distributional effects of the block pricing scheme in São Paulo, Brazil, and found that while rate structures could be designed to be more progressive, these rate structures introduce a trade-off with meeting revenue requirements (i.e., more progressive rates structures may yield less revenue). These findings suggest that IBRs, if not carefully designed, can inadvertently burden larger, potentially lower-income households.

Further empirical findings challenge the assumption that IBRs effectively address equity concerns. Whittington *et al.* (2015) reviewed the empirical literature and noted a general consensus that IBRs do not effectively target subsidies to poor households. Other researchers focusing on energy demand have also pointed out that the correlation between water consumption and income is often weak, undermining the ability of IBRs to serve as an effective redistributive tool (Borenstein 2012; Borenstein and Davis 2012; Levinson and Silva 2022). As Smith (2022) notes in a study of Denver's individualized water rate structure, poorer households often have more members and less efficient homes, which can lead to higher water consumption, resulting in a weak positive correlation between use and income. Consequently, the implicit subsidies provided in the initial blocks of an IBR may not reach the intended low-income households, while non-poor households with low consumption can also benefit, leading to issues of leakage and under-targeting (Nauges and Whittington 2017; Whittington *et al.* 2015; Wichelns 2013).

In contrast to complex IBR structures, there is some recent empirical evidence suggests that simpler rate structures, potentially combined with other mechanisms, may offer more desirable equity outcomes.[17] Wietelman *et al.* (2024) found that

[17]Relatedly, although not equity-focused, there is an active debate about whether consumers understand nonlinear water rates structures (e.g., Shin 1985; Nieswiadomy and Molina 1991; Nataraj and Hanemann 2011; Wichman 2014; Binet *et al.* 2014; Brent and Ward 2019). Regardless of price perception, alleviating affordability burdens relies on reducing *total* expenditures, so the interaction between equity and price perception (e.g., average versus marginal price responsiveness) is seemingly immaterial. However, if consumers do not respond to marginal incentives, then concerns about equity–efficiency trade-offs that arise from discounting marginal prices may be overblown. The role of price misperceptions on equity goals is a worthy avenue for future investigation.

uniform rate structures perform similarly or slightly worse than existing budget-based rates in terms of equity, but pairing a uniform rate with a variable fixed charge tied to income improves the progressivity of the rate structure. This aligns with the theoretical preference for using fixed charges or direct income transfers to address equity without distorting the price signal for water consumption (Levinson and Silva 2022; Wichman 2024). Similarly, Cardoso and Wichman (2022) simulated water affordability policies in the US to illustrate that policies that provide lump-sum rebates to low-income households funded by income taxes achieve affordability targets with fewer distortions compared to policies that change marginal consumption incentives. This, again, suggests that direct assistance programs may be more effective and efficient in addressing water affordability concerns than relying solely on the volumetric price structure. Ultimately, the empirical evidence indicates that the design and effectiveness of rate structures in achieving equity goals is not particularly favorable and that IBRs, while intuitively appealing and commonly adopted, are not an effective instrument for income redistribution or ensuring affordability (Boland 2000; Ruijs *et al.* 2008; Whittington *et al.* 2015).

4.4. *Are customer assistance programs effective*?

Customer assistance programs (CAPs) initiated by water utilities at the local level have the potential to play an important role in ensuring affordable access for low-income households. Cook *et al.* (2020b) is one of the best global reviews on CAPs, providing both a typology of programs as well as a discussion of their effectiveness. Cook *et al.* (2020b) delineate four key elements of CAPs: subsidy administration, funding, targeting and delivery mechanisms. CAPs aim to reduce affordability burdens through various delivery strategies, including connection subsidies to lower initial access costs, consumption subsidies like rebates or bill discounts to directly reduce water expenses, flexible payment terms to make bills more manageable, and conservation assistance to lower usage and thus bills (Cook *et al.* 2020b). Many types of CAPs, especially consumption subsidies, lower the cost of water service, which has the potential to backfire by increasing water usage since these programs can dilute any conservation signal that prices send. A key distinction exists between low- and middle-income countries (LMICs), where the primary focus has been on expanding access for the poor, and wealthy countries, which have increasingly emphasized improving program enrollment, ensuring long-term financial sustainability, and better integrating CAPs with broader social safety nets (Cook *et al.* 2020b).

Despite significant efforts dedicated to designing and targeting CAPs, there is a remarkable lack of empirical evidence regarding their overall effectiveness in alleviating affordability burdens and influencing water use in both LMICs and wealthy nations. Although numerous studies have examined the precision of targeting methods, Cook *et al.* (2020b) note that few have rigorously evaluated the actual impact of CAPs on reductions in affordability burdens, household water demand, or on the cost-effectiveness of the programs. The authors underscore the critical need for more research that moves beyond assessing targeting accuracy to thoroughly evaluate the real-world consequences of CAPs on water access, affordability, and consumption patterns. Ultimately, determining whether these programs “work” to improve the lives of vulnerable populations and promote sustainable water use requires further analysis of behavioral changes and overall program efficacy.

Studies of CAPs in LMICs focus on a variety of approaches to reducing affordability burdens, though evidence on their effectiveness is often limited (Cook *et al.* 2020b). These programs include connection subsidies to help households afford initial access to piped water networks (Cook *et al.* 2020b; Mason 2009), and service-level targeting such as subsidized public taps or kiosks which aim to provide affordable access for the poorest (Cook *et al.* 2020b; Komives *et al.* 2005; Mason 2009). While means-testing is used in some contexts, administrative capacity can be a significant barrier (Cook *et al.* 2020b; Contreras *et al.* 2018). Other targeting mechanisms include proxy measures based on dwelling type or geographic location (Banerjee and Morella 2011; Cook *et al.* 2020b). The effectiveness of these CAPs is varied and influenced by factors like existing infrastructure and the ability to accurately target and deliver assistance to the most vulnerable populations. Rigorous, independent evaluations of the impact of non-tariff CAPs on affordability in LMICs remain scarce.

There is also limited evidence of the effectiveness of CAPs in developed countries, although various case studies provide valuable insights into their design and implementation (Cook *et al.* 2020b). The Tiered Assistance Program (TAP) in Philadelphia, PA, as detailed by Mack *et al.* (2020), offers a concrete example of an innovative income-based billing program designed to reduce affordability burdens. TAP is a means-tested program where income-qualified recipients pay a fixed percentage of their monthly income rather than a water bill based on usage. Early evidence suggests that TAP has generated meaningful enrollment, reaching over 15,000 customers and surpassing the participation rates of previous programs (Mack *et al.* 2020). Participants have reported that TAP helped with their monthly budgets, and the program is funded through a modest surcharge on non-recipient bills. However, TAP has only reached a fraction of eligible households, with efforts to streamline the application process and reduce administrative burdens underway.[18]

[18]See, e.g., https://www.phila.gov/media/20230526113411/Tiered-Assistance-Program-TAP-2022-annual-report.pdf (last accessed: March 21, 2025).

Further, it is unclear whether TAP has had a meaningful, sustained impact on increasing payments, reducing delinquencies, and/or changing water use among participants.

Targeted water affordability programs can be effective, although their success is contingent on program design. Research indicates that lump-sum rebates directed at low-income households, financed through income taxes on non-assisted households, show promise in achieving affordability goals with fewer adverse effects on household spending and water conservation compared to policies that change marginal consumption incentives (Cardoso and Wichman 2022). However, reducing the marginal price of water for low-income customers via “lifeline” rates may be inefficient, and there is a scarcity of empirical evidence on the actual effectiveness of these program types.

CAPs are often funded by utility revenues, sometimes in partnership with local agencies or non-profits, and may utilize means-testing or focus on vulnerable populations (Cook *et al.* 2020b). However, funding these programs requires some level of income redistribution within the utility service area, which may be politically challenging to implement and in some jurisdictions, income redistribution through the water bill is illegal. Additionally, poor communities may not have the ability to cross-subsidize water services if the majority of their service population is poor. This challenge indicates a role for states and federal governments in addressing water affordability. In the US, the federal government could establish a permanent low-income water assistance program modeled after the temporary Low-Income Household Water Assistance Program (LIHWAP) or the Low-Income Household Energy Assistance Program (LIHEAP) by providing grants to states, territories, and Tribes to fund programs that assist low-income households with water and wastewater bills (EPA 2024). Eligibility could be determined based on household income at or below a certain threshold of the federal poverty guidelines or regional incomes. Programs could offer direct financial assistance, bill discounts, or lump-sum rebates to eligible households. Using categorical eligibility for households already enrolled in other means-tested programs (e.g., the Supplemental Nutrition Assistance Program (SNAP) or Supplemental Security Income (SSI)) could streamline enrollment.

5. Concluding Thoughts

Overall, this review establishes several key facts about equity and affordability in residential water service:

- Water affordability is an increasing concern, both in developed and developing nations.

- Water bills are regressive, with lower-income households spending a larger proportion of their income on water services compared to higher-income households.
- Water demand is generally inelastic with respect to income, meaning that as household income changes, water consumption changes by a proportionally smaller amount. However, the relationship between household size and income complicates this relationship.
- Water price responsiveness is generally higher among lower-income households, although higher-income households can also exhibit relatively high price sensitivity under certain conditions.
- Non-price conservation measures, such as outdoor watering restrictions, tend to elicit similar behavioral responses across income groups, but the distributional consequences can differ due to varying consumption patterns and capacities for conservation.
- Water utilities often use increasing block rates (IBRs) to achieve equity goals, but these structures generate allocative efficiency costs and can be ineffective for (or exacerbate) equity goals if not carefully designed, as consumption is a poor proxy for income.
- Direct transfers or progressive fixed fees are often more effective instruments for addressing equity concerns without distorting the price signal for water consumption.
- Customer assistance programs (CAPs) can help alleviate affordability burdens, but their effectiveness varies widely depending on program design and implementation. There is a need for more rigorous evaluations of CAPs to assess their effectiveness.
- There is a growing recognition of the need for national or state water affordability programs in developed countries like the US, similar to existing programs for energy assistance, to ensure that low-income households have access to affordable water services.

Despite these fairly explicit conclusions, there are many gaps in our understanding of equity and affordability for water use that will be necessary to fill before we can implement effective solutions to this growing problem. Although there are thousands of estimates of price and income elasticities of water demand, there are considerably fewer studies that measure affordability burdens using micro-data and that tie household water expenditures with other characteristics, such as income, household size, and demographics. There are also many studies that evaluate increasingly complex water rate designs and their influence on water consumption, but relatively few that evaluate their distributional effects. There are even fewer studies that contrast the pros and cons of simpler, more direct approaches to reducing affordability burdens.

There is a growing set of studies that seek to measure the extent of the water affordability problem in the US (e.g., Teodoro 2018; Cardoso and Wichman 2022; Patterson *et al.* 2023; EPA 2024), many of which use sophisticated approaches to match service territories with demographics or simulate local income distributions to better capture the distribution of water expenditures. While these studies provide important context for the magnitude of the affordability problem, the obvious next step is evaluating whether existing programs are effective at alleviating affordability burdens. Despite the widespread adoption of CAPs across the globe, there is a surprising lack of evidence about whether they actually work. This missing evidence could be a result of the fragmented way in which many water policies are implemented and the hesitation of utilities to share sensitive data on program participation for vulnerable households. However, this information is critical to understanding what types of programs work, how they can be improved, and whether they are cost-effective.

This review so far has not discussed the implications of water affordability issues on non-payment of utility bills, water shutoffs, and tax liens, which can have deleterious consequences for households. Utilities have disparate policies to deal with non-payment and water shutoffs, although there is very little evidence of the effectiveness or long-run implications on household well-being. These issues deserve more attention from researchers.

Finally, there is a need for more research on the distributional consequences of non-price conservation measures, such as outdoor watering restrictions, behavioral nudges, or water-efficiency rebates, which are often implemented during times of drought or scarcity in lieu of using prices to manage demand. While these measures may be effective at reducing overall water use, their impact on different income groups and the potential for unintended consequences (e.g., exacerbating existing inequities) warrant further investigation. Understanding how these policies affect various households can help inform more equitable and effective water management strategies.

To date, the economics literature on residential water has overwhelmingly focused on how to "manage demand" for residential water. However, in any region across the world, residential consumption is a small share of total use and often the highest-value use of water, which makes the focus on residential conservation seem misplaced. Given the increasing concerns about rising infrastructure costs and widespread poverty in both developed and developing countries, there is a growing need to shift the focus of water economics research towards equity and affordability. Indeed, with a significant portion of households facing water affordability burdens

(Cardoso and Wichman 2022; Patterson *et al.* 2023; EPA 2024) and the understanding that current pricing structures may not adequately protect vulnerable populations (Whittington 1992; Komives *et al.* 2005; Wietelman *et al.* 2024), a greater emphasis on the equity dimensions of water pricing and access is becoming increasingly critical.

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